

CS258 Final Project Proposal

RL for Mathematical Reasoning in Small LLMs

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Motivation

We aim to improve mathematical reasoning in *small* language models using reinforcement learning (RL). Most advances in math-capable LLMs rely on scale ($\geq 70\text{B}$ parameters) or specialized math pre-training. We hypothesize that Proximal Policy Optimization (PPO) with dense intermediate rewards can teach reasoning more efficiently by rewarding progress through intermediate solution steps, not just final answers.

We will train Gemma-2-2B-Instruct or Phi 3 3.8B on the GSM8k dataset of grade-school math problems. The baseline model achieves $\sim 25\%$ accuracy; our goal is a 20–30% absolute improvement (45–55% total). Using a general-purpose model rather than a math-specialized one allows a clearer measure of RL-driven gains.

Methodology

We'll use the TRL PPOTrainer to fine-tune Gemma-2-2B under KL-penalized policy updates. The novel element is a mathematical verification environment that checks not only final answers but intermediate reasoning steps.

Each problem elicits a structured solution with Python code and explanations. Executed in a sandbox, the code is compared to the ground truth. Rewards are hierarchical: +0.1 for syntactically valid Python, +0.2 for successful execution, +0.3 for correct numeric extraction, +0.5 for partial correctness, +2.0 for full correctness, -0.5 for incorrect final answers

Training proceeds via curriculum learning:

- Weeks 1–2: easiest 1 k problems, basic reward shaping $\rightarrow$ target +10–15 % gain
- Weeks 3–4: full GSM8k, enhanced step rewards $\rightarrow$ target +20–30 % gain
If successful, we'll extend to the MATH dataset (levels 1–2) or scale to Phi-3.5-mini (3.8B).

Evaluation

We'll evaluate on GSM8k test data using:

- Primary metric: answer accuracy
- Baselines: zero-shot ($\sim 25\%$), few-shot ($\sim 30\text{--}35\%$), RL-trained (target 45–55 %)
- Secondary metrics: sample efficiency, intermediate-step correctness, policy KL stability, and reward-curve smoothness

References

1. Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., & Iwasawa, Y. (2022). *Large Language Models are Zero-Shot Reasoners*. NeurIPS 2022.
2. Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., et al. (2021). *Training Verifiers to Solve Math Word Problems*. arXiv:2110.14168.
3. OpenAI (2022). *Learning to Summarize with Human Feedback*. arXiv:2009.01325.